

Fitness consequences of transposable element dynamics in a small, isolated bird population

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Abstract

The evolutionary consequences of transposable element (TE) dynamics may have a profound effect on small and isolated populations. TEs rapidly generate genomic variation, and new insertion sites can potentially cause large phenotypic effects. As with any other mutation, TE insertions can either be deleterious, neutral, or adaptive. Therefore, differential TE transposition rates can drive various evolutionary processes in small populations, especially when the population has suffered genetic erosion. We explore the consequences of TE dynamics on survival, lifetime reproductive success (LRS), and reproductive senescence in an endangered bird population. Whole-genome sequences allowed us to estimate the genomic TE content, and the proportion of autosomal runs of homozygosity as a measure of inbreeding. We tested for effects of TE dynamics on individual fitness, while controlling for inbreeding depression and other factors known to affect the chosen fitness components. We found evidence consistent with “W-toxicity”, caused by an accumulation of active endogenous retroviral (ERV) elements in the W chromosome, in reducing the survival of female offspring. However, this effect depended on exposure to higher rainfall variance during early life. We also found an association between larger maternal TE contribution and increased male offspring LRS and a marginal negative association between paternal TE contributions and male offspring LRS. Finally, we did not find any

association between TE dynamics and reproductive senescence. Our results therefore suggest a trade-off between female survival and male offspring LRS driven by maternal TE contributions. They also provide, to our knowledge, the first evidence of TE dynamics related fitness effects in a wild vertebrate population.

Introduction

At the microevolutionary scale, evolution is propelled by the appearance of new genetic variants, either through point mutations (single-nucleotide polymorphisms; SNPs), indels (insertion-deletions) or larger scale structural variation, all of which can alter phenotypes and may then be selected for (Agashe et al., 2023). However, the genotype-phenotype map is complex; many regulatory processes translate genetic variation into phenotypic variation, and not all mutations lead to different phenotypes (Agashe et al., 2023; Chevin et al., 2022). In fact, most mutations are neutral and have little effect (Agashe et al., 2023). However, variation arising from alternative mechanisms such as the activity of transposable elements (TEs) can lead to large phenotypic effects (Casacuberta & González, 2013; Lanciano & Cristofari, 2020; Schlenke & Begun, 2004; Slotkin & Martienssen, 2007).

TEs are molecular parasites found in the genome of all living beings (Bourque et al., 2018). They are composed of genetic material and can create new copies of their sequence in other regions of the genome, either by cutting and pasting their sequence or by retrotransposition of an RNA intermediate (Bourque et al., 2018). These modes of replication are the main characteristic that classifies TEs into Class I retrotransposons that copy and paste their sequence with the help of a retrotranscriptase, and Class II DNA that cut and paste their sequences elsewhere in the genome (Bourque et al., 2018).

The mobility of TEs in the genome can affect phenotypes in several ways. First, they can insert themselves within functional genomic regions, causing loss-of-function mutations that can change the

phenotype (Bourque et al., 2018). Second, they can transport cis-regulatory sequences, such as promoter regions, and protein coding regions to other parts of the genome, potentially changing gene expression patterns and producing gene variants across the genome (Bourque et al., 2018; Casacuberta & González, 2013; Lanciano & Cristofari, 2020; Slotkin & Martienssen, 2007). Third, they can increase the genome size, which can affect development and fitness under certain environmental regimes (Ågren & Wright, 2011; Valdebenito-Maturana et al., 2021). Finally, DNA transposon activity has been associated to higher mutation rates around their excision sites due to the activation of DNA repair pathways (Wicker et al., 2016), but DNA transposons are considered generally inactive in avian genomes (Kapusta & Suh, 2017). Thus TEs can have an important role in shaping phenotypic variation that might overshadow the effects of point-mutations (Catlin & Josephs, 2022). These phenotypic effects are often deleterious, as TE expression can lead to genome instability and changes in gene expression, which might alter the regular functioning of the cell (Andrenacci et al., 2020; Horváth et al., 2017; Pappalardo et al., 2021; Slotkin & Martienssen, 2007; Wood & Helfand, 2013). Given the deleterious consequences, gene expression suppression mechanisms might be deployed to silence TE expression, but in doing so, nearby genes may also be downregulated and cause a loss of adequate gene expression (Betancourt et al., 2024; Hollister & Gaut, 2009; Iwasaki et al., 2015; Slotkin & Martienssen, 2007). TEs can (rarely) lead to beneficial genetic variation, potentially even leading to rapid adaptation in novel environments (Casacuberta & González, 2013; González et al., 2010; Schrader et al., 2014; Stapley et al., 2015). For example, industrial melanism in the peppered moth *Biston betularia*, originated from a TE insertion in the *cortex* gene (Hof et al., 2016). However, despite such examples, the expression of TEs in somatic tissues is usually related to disease, senescence and genome instability (Andrenacci et al., 2020; Betancourt et al., 2024; Colombo et al., 2018; Driver & McKechnie, 1992; Pappalardo et al., 2021; Slotkin & Martienssen, 2007; Wood & Helfand, 2013).

TEs are of special interest as sources of genetic variation in scenarios where genetic diversity might be depleted, such as invasive species or endangered populations (De Kort et al., 2022; Schrader et al., 2014; Stapley et al., 2015). TEs have been found to contribute to the rapid adaptation of invasive

species by contributing to genetic diversity through expression bursts linked to TE islands (Schrader et al., 2014; Stapley et al., 2015). In endangered populations, isolation and low population size can lead to inbreeding, elevated genetic drift and genetic erosion, causing inbreeding depression and a loss of adaptive potential (Frankham et al., 2019; Nordstrom et al., 2023; L. I. Wright et al., 2008). In turn, these factors lead to lower individual and population fitness and thus a higher risk of extinction (Frankham et al., 2019; Pemberton et al., 2017; L. I. Wright et al., 2008). In such situations TE insertions and their effects on cis-regulation and mutagenesis around their insertion sites may provide vital genetic diversity to counteract inbreeding depression, as suggested in a study comparing the role of TEs in the evolution and survival of selfing and outbreeding populations of *Arabidopsis lyrata* (De Kort et al., 2022). They observe genomic heterozygosity around TE insertions, and extends further than the expected distance in which mutation rate is expected to increase around DNA transposons (Wicker et al., 2016), so they suggest that what maintains heterozygosity is the preference of TEs to insert near house-keeping genes, maintaining their diversity either by exon shuffling or mutagenesis (De Kort et al., 2022).

Unravelling the evolutionary consequences of TE dynamics is difficult due to the varying effects of TE activity on fitness and phenotypic variation (Bourque et al., 2018; Catlin & Josephs, 2022). In avian genomes, active copies of endogenous retroviruses (ERVs) belonging to the order of long terminal repeat (LTR) elements are shown to accumulate in the W chromosome, where they escape post-transcriptional suppression and epigenetic silencing (Peona et al., 2021; Warmuth et al., 2022). This accumulation of ERVs in the W chromosome has been posited as a driving aspect of TE dynamics in avian genomes, potentially leading to a reduction in female lifespan through the expression of these ERVs in late life, analogous to the “Y toxicity” hypothesis observed in *Drosophila* (Brown et al., 2020). The “W toxicity” proposed for avian genomes could also be the underlying cause of female biased mutational load and genetic incompatibilities (Peona et al., 2021).

Capturing the potential trade-offs between the positive and negative potential consequences of TEs in a wild population requires detailed data on both individual variation in fitness and in genomic TE content. The long term, intensive monitoring of the Seychelles warbler (Pti' Merl de Zil) *Acrocephalus sechellensis*, population on Cousin Island presents an ideal study system to understand the evolutionary consequences of varying TE dynamics in populations affected by inbreeding. Previous studies on this passerine have found the species to lack any effective inter-island dispersal (Komdeur et al., 2004), that the species experiences actuarial and reproductive senescence (Hammers et al., 2019; Sparks et al., 2021; Tsui et al., 2026), and have shown evidence of the population experiencing inbreeding depression after a population bottleneck that reduced the effective population size to <50 (Lee, et al., 2026b; Richardson et al., 2004; Spurgin et al., 2014). Recent work has managed to create a valuable genomic toolkit which has been leveraged to assemble a new reference genome, refine the population pedigree, get cutting-edge estimates of inbreeding in the population and use a genomic approach to assess inbreeding depression (Lee, et al., 2026a; Lee, et al., 2026b). The genomic resources for the species also allowed us to confirm varying transposition rates in individuals in this population (González-Mollinedo et al., Chapter 2). Varying transposition rates in a population experiencing inbreeding depression, and the increase in TE activity under stressful conditions, may be mechanisms through which TEs increase heterozygosity and possibly counteract the detrimental effects of inbreeding depression as in *A. lyrata* (De Kort et al., 2022), but this mechanism has not been observed in any vertebrate populations to date.

In this study, we investigate associations between TE dynamics and survival, lifetime reproductive success (LRS), and reproductive senescence. With these models, we will try to differentiate between fitness variation associated to TE dynamics and the detrimental effects of inbreeding depression, expecting to find evidence of TEs contributing to counteract inbreeding depression. However, we expect to find trade-offs between any positive fitness associations and TE activity, such as decreased female survival and possible changes in the trajectory of reproductive senescence associated to the TE load found in the W chromosome (Peona et al., 2021). These possible

outcomes of TE activity in the Seychelles warbler could lead to clearly distinguishable consequences in survival, reproductive success and senescence, and possible implications on the role of TEs in the evolvability of inbred populations.

Methods

Study system

This study focused on the consequences of TE dynamics in the population of the Seychelles warbler (Pti' Merl de Zil) on Cousin Island, Seychelles (04°20'S, 55°40'E). The Seychelles warbler is a facultatively cooperative breeding passerine bird (Hammers et al., 2019; Komdeur et al., 2006). Once probably widespread across the Seychelles islands before human colonization (Spurgin et al 2014) by the 1900's it was limited to Cousin Island, where it reached a minimum effective population size of <50 (Spurgin et al., 2014). The population has since recovered from this bottleneck and new populations have been established through translocation in Fregate, Denis, Cousine and Aride islands (Komdeur, 1994; D. J. Wright et al., 2014)

The population on Cousin has been monitored consistently since 1985 (and to a greater extent since 1997 (Borger et al., 2023; Lee, Bartleet-Cross, et al., 2026), with different aspects of the life history of practically every bird in the island recorded. Birds are caught as nestlings or within 3 months of fledging using mist nets (Hammers et al., 2019; Kingma et al., 2016). Each bird is assigned a unique combination of three color rings and a metal ring with a unique code from the British Trust of Ornithology (BTO). Birds are aged using a combination of observations on nesting/fledgling and subsequent behavior (e.g. begging), or for older individuals on eye color (Komdeur, 1991). Around 115 territories are present in the island (Borger et al., 2023), and individuals can remain in their natal territories as subordinates or helpers (Hammers et al., 2019; Kingma et al., 2016; Komdeur et al., 2006). When sexually mature, individuals can dispute breeding positions or disperse to occupy the breeding position in other territories where group composition may vary (Brouwer et al., 2006; Eikenaar et al., 2008; Kingma et al., 2016). A blood sample of ~25-50 μ L is obtained from captured

individuals for molecular sexing (Griffiths et al., 1996; Sparks et al., 2021), parentage assignment (Lee, et al., 2026a; Sparks et al., 2021) and genomic sequencing (Lee et al., 2026a).

Observational data on group behavior and nesting is collected across the island during both breeding seasons, although predominantly during the major breeding season between June-October (Komdeur & Daan, 2005), providing a census of the population and allowing determination of who is resident on the ca. 115 territories on the island (Borger et al., 2023). Resighting rates are >98% across the island and allows to assign and track the social roles of the observed birds in each group (Brouwer et al., 2006). Parentage assignments are done using molecular methods but also corroborated with behavioral observation, leading to very accurate parentage assignments across the population and the quantification of extra pair paternity (EPP) accounted in the generated pedigree (Hadfield et al., 2006; Richardson et al., 2001; Sparks et al., 2021, 2022). Clutch sizes are of one egg around 80% across nesting, and the breeding pair will tend for the nestling with the assistance of helpers when present (Hammers et al., 2019; Komdeur et al., 2006). Additionally, these group observations and the high resighting rate in the population allow for extremely detailed estimates of lifespan and lifetime reproductive success (Brouwer et al., 2006).

TE content and inbreeding coefficients

Whole-genome sequences were obtained from 1880 individuals as part of a collaboration with another study (Lee, et al., 2026a). DNA was extracted either by using an ammonium acetate extraction protocol (Richardson et al., 2004) or using DNEasy kits (Qiagen, Crawley, UK). Libraries were prepared at the Centre for Genomic Studies in Liverpool. Samples were submitted in 20 randomly selected batches and libraries were prepared using NEBNext Ultra II FS DNA LibraryPrep Kit (New England Biolabs, ~300 bp inserts) on the Mosquito platform using a 1/10 reduced volume protocol. Paired-end sequencing was performed on a NovaSeq-6000 (Illumina). Short-read (150 bp) whole-genome FASTQ files were obtained at low coverage (mean = 2.65x, SD = 1.70x). A set of 56 unrelated

individuals was resequenced to increase coverage (mean = 14.22×, SD = 2.30×) to use as a validation set in down sampling verification of the accuracy of genotype imputation.

Genomic data was verified to their samples of origin and then processed for SNP calling. Read adapters were trimmed and filtered by quality with a Phred score of 33 using *Trimmomatic* 0.39-Java-11 (Bolger et al., 2014). The trimmed reads were aligned to the reference genome obtained by Lee et al. (2026a) using *BWA-MEM* 0.7.17-GCCore-12.2.0 (Li & Durbin, 2009). Resulting alignment files were cleaned from unmapped reads using *SAMtools* v 1.19.2-GCC-13.2.0 (Li et al., 2009). SNPs were called using *mpileup* and *call* functions in *BCFtools* v 1.19-GCC-13.2.0, filtering out positions by mapping quality (-q) and base quality (-Q) < 20 and a prior for expected substitution rate (-P) of 1x10⁻⁶.

Given the low coverage recovered for the whole-genome sequence files, imputation was performed with *STITCH* 1.7.0 (Davies et al., 2016), obtaining an imputation accuracy between 88.7-97.6% across the chromosomes in the genome assembly (Lee, et al., 2026). Long runs of homozygosity (ROH) were obtained using *PLINK* 1.90b7 (Purcell et al., 2007). Long ROHs were defined to have a minimum length of 3.8 Mbp as a cut off, as this length represented long ROHs in the Seychelles warbler, where it has been shown to be associated with inbreeding depression (Lee et al., 2026b), and was the *PLINK* parameters used for the *homozyg* function were --geno 0.05 --homozyg-density 200 --homozyg-gap 300 --homozyg-het 2 --homozyg-kb 1000 --homozyg-snp 50 --homozyg-window-het 2 --homozyg-window-missing 2 --homozyg-window-snp 50 --mind.

TE content was determined using *dnaPipeTE* 1.4c (Goubert et al., 2015; Goubert et al., 2023), which assembles a *de novo* repeatome per sample using raw next-generation-sequencing reads. *dnaPipeTE* is designed to handle low coverage sequencing (<0.1X). As mitochondrial reads can be mislabeled as TEs by *dnaPipeTE*, we first excluded reads aligned to the mitochondrial genome of the reed warbler, *Acrocephalus scirpaceus* (Singh et al., 2008), using *BWA-MEM* 0.7.17-GCCore-12.2.0 (Li & Durbin, 2009) on trimmed reads and recovering FASTQ files that were used as input using *SAMtools*. We used three “draws” of 0.1× genome coverage per sample, which were then passed through a

normalization step, ensuring that no more than 50 reads per base would be taken into account, allowing for well-distributed reads across the genome. Default annotation thresholds in *dnaPipeTE* were used. A custom TE library for the Seychelles warbler reference genome was used to identify TEs with at least 20% of the target sequence mapped by reads (González-Mollinedo et al., Chapter 2). The aligned base-pairs were added up and considered as the proportion of an individual genome consisting of TEs.

The log-ratio between parents and offspring were considered as parental TE contributions to offspring, treating a fraction of parental TEs as active and contributing to offspring genomic TE content (GTEC), similarly to compositional data (Greenacre, 2021). Parent-offspring trios were obtained from a pedigree (1992-2022), derived using up to 30 microsatellite markers (Hadfield et al., 2006; Richardson et al., 2000; Sparks et al., 2021), and corrected with whole-genome data (Lee, et al., 2026a). An increase in GTEC was considered to reflect novel TE insertions, especially in the germline, while a decrease in GTEC is thought to be caused by the disruption of their sequence due to recombination or unsuccessful detection (Betancourt et al., 2024). Given that we lack TE expression data, this approach allowed us to obtain a genome-wide proxy of TE activity in the parental germline reflected in the differences between parent TE content and offspring TE content, assuming that positive contributions occurred through transposition (González-Mollinedo et al., Chapter 2).

Statistical analyses

Our goal was to understand the effects of individual genomic TE content and parental TE contributions on survival, lifetime reproductive success (LRS) and reproductive senescence. We controlled for factors that have already been shown to impact fitness components and senescence in the Seychelles warbler. This included how inbred the focal individual or its parents were (F_{ROH}) (Kyriazis et al., 2025; Lee et al., 2026b), the presence of helpers in the territory during offspring development (Hammers et al., 2019; Komdeur et al., 2006; Paquet et al., 2015), individual and maternal age at conception (Richardson et al., 2004; Sparks et al., 2021, 2022). We also included

measures of environmental stress (the variance in precipitation experienced) by a parent before offspring hatching, and by the offspring during early life, as stress could be linked to the activity of TEs and other fitness components (Fox & Reed, 2011; Horváth et al., 2017; Madlung & Comai, 2004, González-Mollinedo et al., Chapter 2). The combination of variables included in specific analyses are given in Table 1. All continuous variables were scaled (i.e. mean-centered and divided by the standard deviation). All models were run in R 4.2.2 (R Core Team, 2022).

Survival

Only individuals who survived the first year of life, were confirmed dead, and where both parents were assigned in the consensus pedigree and had WGS data, were included (N=291 focal individuals; 149 male and 142 female). We ran a Cox proportional hazards model with mixed effects using the package *coxme* 2.2-18.1 (Therneau, 2022) (see Table 1). We additionally tested for sex-biased effects, mother's age at hatching, and focal GTEC, and the interaction between parental TE contributions and environmental stress in early life. Year was included as a fixed effect to account for a bias towards early deaths caused by including more recent death events in our dataset (Bouwhuis et al., 2015). We controlled for the random effects of bird identity, father identity, mother identity and birth year to account for pseudo-replication and cohort effects. Significant sex-specific interactions were then evaluated in sex-specific models to assess the significance and direction of the effects.

Lifetime reproductive success (LRS)

We ran a zero-inflated model with a Poisson distribution on LRS to account for the instances where LRS was 0. We included individuals with both parents in the pedigree, who were dead by the time of the analysis, whose reproductive success was verified (including EPP), and where genomic data was available for them and both parents for measures of inbreeding and TE content. We controlled for the random effects of identity of the father and mother, and birth year to account for pseudo-replication

and cohort effects. For all variables see Table 1. The model was run using glmmTMB 1.1.9 (Brooks et al., 2017).

Reproductive senescence models

We ran a zero-inflated generalized linear mixed model with a generalized Poisson distribution on annual reproductive success. Generalized Poisson models account for underdispersion, which was observed for annual female reproductive success. The model was run on 1028 yearly observations of 97 birds, filtering the individuals used in the LRS model to those where group composition was known to determine yearly helper presence following (Sparks et al., 2021). We used a similar model to LRS, but mean-centered age to assess within-individual age effects to test for differences in senescence between individuals (van de Pol & Wright, 2009). Reproductive senescence models were run separately for males and females to maintain the assumption of homoscedasticity, as the sexes showed different variances in their reproductive output (Sparks et al., 2022) in our dataset (Levene's test for homogeneity of variance p -value < 0.0001). These models were run using glmmTMB 1.1.9 (Brooks et al., 2017). We controlled for the random effect of bird identity to make sure that we obtained individual senescence trajectories, as well as maternal and paternal IDs to account for pseudo-replication.

Model checks

We confirmed that the distribution of model residuals was normal using DHARMA 0.4.6 (Hartig, 2022), we discarded singular fits and checked for multicollinearity (all models kept only variables with VIF < 3), and discarded overdispersion using performance 0.10.8 (Lüdecke et al., 2021) for all our models. Non-significant interactions were removed. This led to some modifications in our models, like excluding individual GTEC from our LRS model since it had a VIF > 5, and we considered the variance inflation come from the correlation between offspring GTEC and parental TE contributions, so only parental proportions were included. Our LRS and reproductive senescence models had singular fits, so we removed paternal ID as a random effect from our LRS model as it contributed very minimally to

276 the explained variance of the model. Both parental ID's were removed as random effects from the
277 reproductive senescence model as it also contributed very little to the explained variance of the model
278 and only a few siblings were present in the dataset. Bird ID was the main explanatory random effect
279 in the senescence model and allowed to group within-individual annual reproductive events. The
280 removal of random effects resolved singular fits and changed p-values slightly, but all significant
281 factors in the models remained the same, showing that final model structure was adequate to assess
282 effect sizes and significance.

283 **Table 1.** Variables considered in the survival, LRS, and senescence models. Random factors are also included in the description of each model. Interaction
 284 effects considered in our models were tested between the pool of variables that made sense to explore as interactions.

Factor	Description	Survival	LRS	Senescence
Mother TE contribution	Difference between focal and mother TE content	x	x	x
Father TE contribution	Difference between focal and father TE content	x	x	x
Focal individual genomic TE content	Proportion of the genome annotated as TEs	x	x	x
Focal individual inbreeding coefficient	Proportion of the genome found in runs of homozygosity of the focal individual	x	x	x
Father's inbreeding coefficient	Proportion of the genome found in runs of homozygosity of the father	x	x	x
Mother's inbreeding coefficient	Proportion of the genome found in runs of homozygosity of the mother	x	x	x
Mother's age	Age of the mother at hatch of the focal individual	x		
Helper presence in natal territory	Presence of helpers in the natal territory when the focal individual hatched	x	x	
Helper presence in the territory	Presence of helpers in a given year (Sparks et al., 2021)			x
Age	Age of the focal individual	x		
Mean-centered age	Within-individual mean-centered age of reproduction			x
δ age	Within-individual displacement from mean age of reproduction			x
δ age ^2	Quadratic term of δ age (displacement from mean age of reproduction)			x
Sex	Sex of the focal individual	x	x	
Rainfall variance in early life	Variance in rainfall from hatching to age of blood sampling (<5 months old)	x	x	x
Maternal rainfall variance	Variance in rainfall experienced from maternal hatching until offspring hatching	x	x	x
Paternal rainfall variance	Variance in rainfall experienced from paternal hatching until offspring hatching	x	x	x
Year	Year the observation was taken (censor data)	x		x
Hatch year	Year when the egg containing the focal individual was hatched		x	
Lifespan	Lifespan in years of the focal individual			x
Random effects	Description	Survival	LRS	Senescence
Year	Year the data were taken	x		
Lay Year	Year when the focal individual hatched			
Bird ID	Focal individual Bird ID generated during monitoring	x		x
Mother ID	Mother individual Bird ID generated during monitoring	x	x	
Father ID	Father individual Bird ID generated during monitoring	x		

285

Results

Survival

No significant effects of individual GTEC or parental TE contribution on survival were detected, but survival was negatively associated with the focal individual's inbreeding coefficient and the mother's age (Cox hazard ratio >1; Table 2, Figure S1). In contrast, rainfall variance in early life, and paternal inbreeding had significant Cox hazard ratios <1, suggesting a positive association with survival. There were also significant interactions between offspring sex and mother's age, and offspring sex and individual GTEC.

Post-hoc sex-specific models showed GTEC was not linked to survival for either sex but mother's age was linked to lower survival for females (Table S1; Figure 1A). They also showed that the significant positive effect of father's inbreeding on survival only occurs in male offspring, despite no significant interaction between sex and father's inbreeding in the main model (Table S2, Figure 1B). As expected, age and year of observation were significantly linked to reduced survival, with the latter accounting for the bias present by the inclusion of birds hatched more recently having shorter lifespans. Additionally, the female-specific model showed a significant interaction between maternal TE contributions and rainfall variance experienced during early life as significantly reducing survival in female offspring (Table 3, Figure 1A).

Table 2. A Cox proportional hazards mixed effect model of the effect of estimated whole genome transposable element (TE) dynamics on the survival of Seychelles warblers on Cousin Island. Regression coefficients, hazard ratios, standard errors (SE), Z-values and p-values are reported. Statistically significant hazard ratios have standard errors that do not overlap with 1 (p-values < 0.05), shown in bold. Hazard ratios <1 indicate a factor that improves survival, while a hazard ratio >1 indicates a decrease in survival.

Factor	Coefficient	Hazard Ratio	SE	Z-value	P-value
Mother TE contribution	0.072	1.074	0.118	0.610	0.540
Father TE contribution	0.002	1.002	0.109	0.020	0.980
Offspring genomic TE content (GTEC)	-0.004	0.996	0.154	-0.030	0.980
Rainfall variance	-0.189	0.828	0.088	-2.150	0.031
Maternal rainfall variance	-0.145	0.865	0.121	-1.190	0.230
Paternal rainfall variance	0.035	1.035	0.114	0.310	0.760

Inbreeding coefficient	0.202	1.223	0.070	2.890	0.004
Mother inbreeding coefficient	0.118	1.125	0.072	1.650	0.100
Father inbreeding coefficient	-0.177	0.838	0.073	-2.420	0.016
Mother's age	0.302	1.352	0.102	2.950	0.003
Helper presence in natal territory	-0.049	0.953	0.152	-0.320	0.750
Age	0.095	1.100	0.023	4.210	<0.0001
Sex (male)	0.125	1.133	0.141	0.890	0.380
Period year	0.068	1.070	0.021	3.300	0.001
GTEC:Sex (male)	0.292	1.339	0.136	2.150	0.032
Sex (male):Mother's age	-0.360	0.697	0.142	-2.540	0.011
Random effects	Variable	Levels	SD	Variance	
Period year	Intercept	25	0.466	0.217	
Bird ID	Intercept	291	0.435	0.189	
Mother ID	Intercept	169	0.312	0.097	
Father ID	Intercept	163	0.020	0.000	

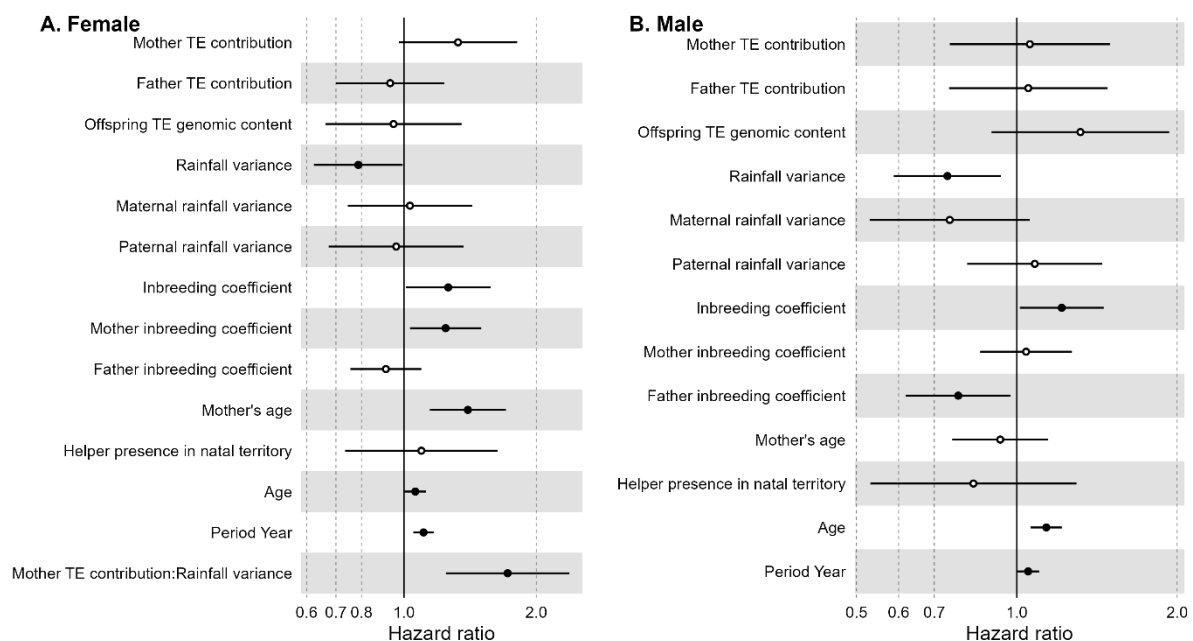


Figure 1. Hazard ratios from sex-specific post-hoc Cox proportional hazards models with mixed effects testing for associations with survival. Point estimates of hazard ratios with 95% confidence intervals are shown. Significant hazard effects are shown with a filled point.

Lifetime Reproductive Success (LRS)

There was a strongly significant interaction between the mother's TE contribution and sex on offspring LRS (Table 3, Figure 2A); i.e. an increase in male offspring LRS with higher maternal TE contributions

but no effect of male offspring LRS. In contrast there was a marginally significant (p-value <0.1) for an interaction between the father's TE contributions and sex on offspring LRS (Table 3, Figure 2A); male offspring LRS decreased with higher paternal TE contributions (Figure 2B) but there was no effect on female offspring LRS. Additionally, there was a significant negative association between inbreeding coefficient and LRS (Table 3, Figure 2C). Finally, a significant negative association between hatch year and LRS accounted for the reduced LRS resulting from the inclusion of more recently hatched individuals who had a shorter lifespan to reproduce.

Table 3. Summary of a zero-inflated Poisson model exploring associations between transposable element (TE) dynamics and other factors with lifetime reproductive success in the Seychelles warbler on Cousin island. Regression coefficients, standard errors (SE), Z-values and p-values are reported. Significant p-values (< 0.05) are shown in bold.

Factor	Coefficient	SE	Z-value	P-value
Intercept	106.564	31.232	3.412	0.001
Mother TE contribution	-0.158	0.110	-1.430	0.153
Father TE contribution	-0.060	0.116	-0.518	0.604
Rainfall variance	-0.058	0.072	-0.800	0.424
Maternal rainfall variance	-0.049	0.117	-0.414	0.679
Paternal rainfall variance	0.142	0.112	1.266	0.205
Inbreeding coefficient	-0.211	0.066	-3.197	0.001
Mother inbreeding coefficient	-0.010	0.080	-0.125	0.901
Father inbreeding coefficient	-0.017	0.070	-0.237	0.813
Sex	0.142	0.127	1.124	0.261
Mother's age	-0.036	0.067	-0.535	0.593
Hatch year	-0.053	0.016	-3.372	0.001
Helper presence in natal territory	-0.078	0.146	-0.535	0.593
Mother TE contribution:Sex	0.504	0.169	2.987	0.003
Father TE contribution:Sex	-0.338	0.175	-1.926	0.054
Random effects	Variable	Levels	SD	Variance
Mother ID	Intercept	89	0.493	0.243

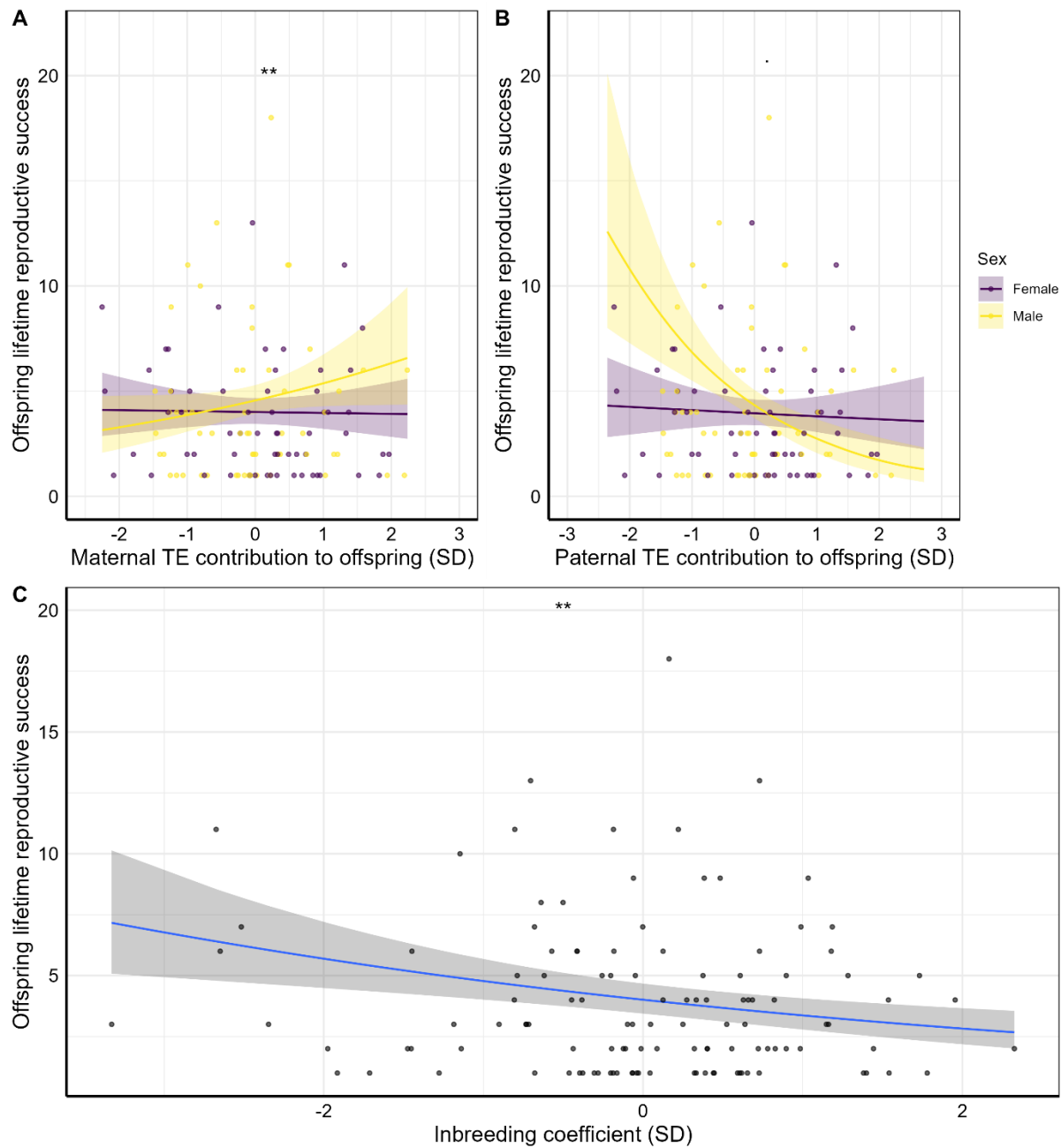


Figure 2. Significant associations with the lifetime reproductive success (LRS) in the Seychelles warbler population on Cousin island with (A) maternal TE contributions (B), paternal TE contributions and (C) inbreeding coefficients. The variables shown in the x-axis have been scaled and are shown in standard deviations (SD). Model predictions are shown as plotted lines, with a 95% confidence interval shown as shaded areas. Points represent the raw data used in the model and significance levels are at the top of each panel.

Table S3. Summary of the sex-specific reproductive senescence models run to estimate the effects of parent TE contributions and inbreeding coefficients on focal senescence trajectories in Seychelles warblers. These models are mean-centered at the individual men age of breeding, and age is measured as a within-individual displacement from mean age (δ). Reproductive senescence is confirmed for both sexes. Male reproductive senescence showed an significant interaction with maternal inbreeding coefficients, showing an earlier onset of senescence in male offspring to be associated with mothers with higher inbreeding coefficients. Regression coefficient, standard error (SE), Z-values and p-values are reported. Significant p-values (< 0.05) are shown in bold.

Male senescence model					Female senescence model				
Factor	Coefficient	SE	Z-value	P-value	Factor	Coefficient	SE	Z-value	P-value
Intercept	-0.429	0.256	-1.677	0.094	Intercept	-0.358	0.298	-1.204	0.229
Mother TE contribution	0.194	0.109	1.787	0.074	Mother TE contribution	0.074	0.126	0.590	0.555
Father TE contribution	-0.062	0.104	-0.592	0.554	Father TE contribution	-0.078	0.143	-0.545	0.586
Rainfall variance	-0.022	0.074	-0.293	0.769	Rainfall variance	-0.059	0.112	-0.533	0.594
Maternal rainfall variance	0.119	0.117	1.019	0.308	Maternal rainfall variance	0.013	0.148	0.086	0.932
Paternal rainfall variance	-0.059	0.121	-0.490	0.624	Paternal rainfall variance	-0.032	0.174	-0.182	0.855
δage	-0.108	0.022	-4.886	<0.0001	δage	-0.213	0.021	-10.019	<0.0001
δage ^2	-0.019	0.007	-2.762	0.006	δage ^2	-0.024	0.006	-4.128	<0.0001
Mean age of reproduction	0.083	0.046	1.795	0.073	Mean age of reproduction	0.012	0.054	0.214	0.830
Inbreeding coefficient	-0.131	0.073	-1.801	0.072	Inbreeding coefficient	-0.058	0.106	-0.545	0.585
Mother inbreeding coefficient	0.094	0.093	1.002	0.316	Mother inbreeding coefficient	0.143	0.098	1.460	0.144
Father inbreeding coefficient	-0.054	0.088	-0.607	0.544	Father inbreeding coefficient	0.100	0.101	0.987	0.324
Helper presence	0.122	0.065	1.870	0.061	Helper presence	0.047	0.050	0.932	0.351
Group size	-0.015	0.061	-0.247	0.805	Group size	0.015	0.062	0.238	0.812
δage:Mother inbreeding	-0.053	0.022	-2.432	0.015					
Random effects	Variable	Levels	SD	Variance	Random effects	Variable	Levels	SD	Variance
Bird ID	Intercept	49	0.277	0.076	Bird ID	Intercept	48	0.482	0.233

Reproductive senescence

We found no associations between observed TE dynamics and annual reproductive success or reproductive senescence in either female or male Seychelles warblers (Table 4). However, we did find a significant interaction effect between mother's inbreeding and age of male offspring (Table 4), indicating that higher maternal inbreeding coefficients were associated with an earlier onset and higher rate of reproductive senescence in male offspring (Figure 3).

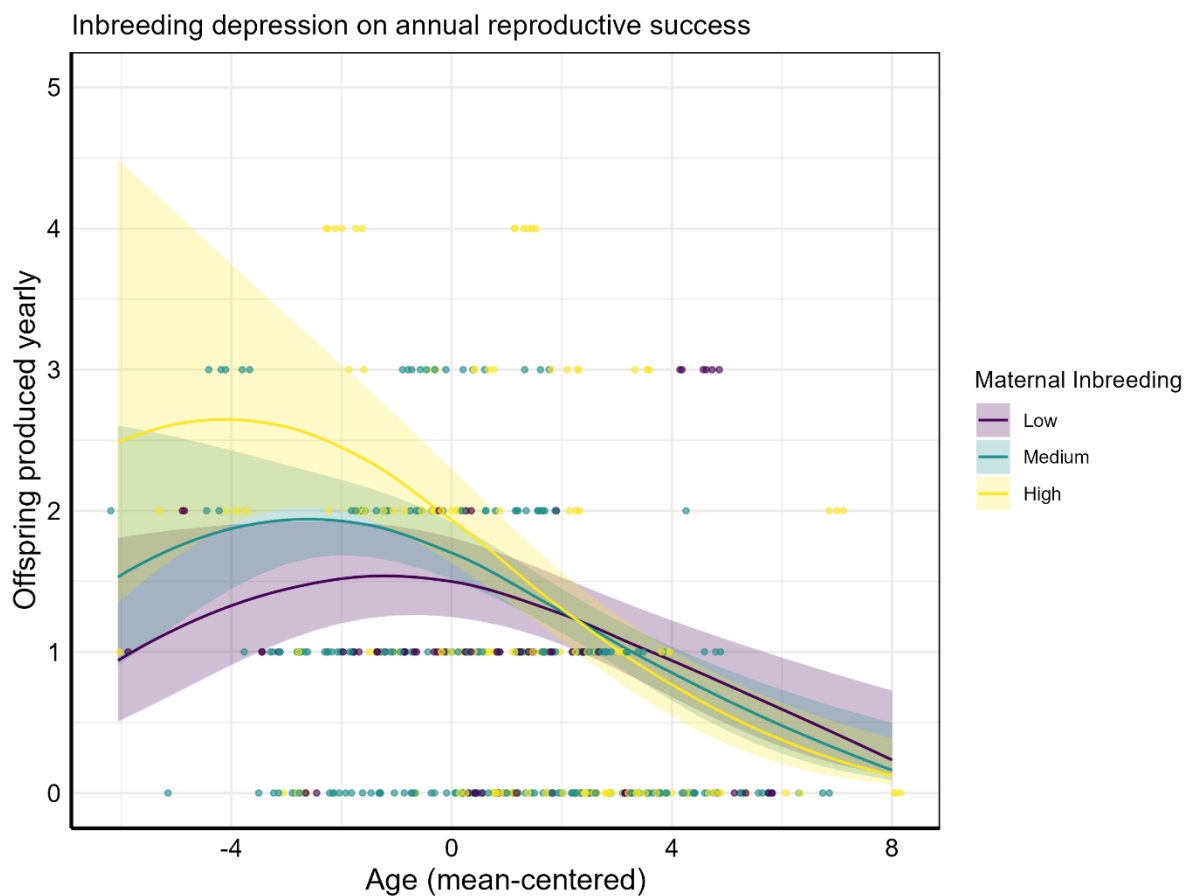


Figure 3. Association between maternal inbreeding levels and male reproductive senescence. Each curve follows the model prediction for a different level of maternal inbreeding, showing the model fit (line) and the 95% confidence intervals around the prediction. Raw data is shown and classified into each level of inbreeding.

Discussion

In this study, we investigated the effects that TE dynamics can have on different fitness components such as survival, LRS and reproductive senescence in a small and isolated population of birds. Our analyses showed sex-specific associations of TE dynamics with survival and LRS, but not with reproductive senescence. Female offspring showed a negative association between survival and maternal TE contributions exacerbated by exposure to higher rainfall variance during early life (Figure 1A). A positive association between male offspring LRS and maternal TE contribution was found as well (Table 3, Figure 2A). These associations pose a trade-off around maternal TE contributions and the sex of offspring.

Sex-specific effects of TE dynamics in survival

We found significant interactions between offspring sex and individual GTEC and mother's age, and so constructed post-hoc sex-specific survival models to better understand our results (Figure 1). The initial interaction we found between GTEC and sex disappeared, but an interaction between maternal TE interaction and rainfall variance during early life was significant in the female survival model. Given the mean-centered scaling of the data in the model, we can assume that above-average values of rainfall variance exposure or maternal TE contributions are positive, while below-average values will be negative. The interaction term tells us that above-average rainfall variance and maternal TE contributions act in tandem as a hazard, while below-average values of any of the variables could override the negative association of the other. Therefore, this interaction shows that the negative association of maternal TE contributions to female offspring survival will be modulated by the exposure to environmental stress in early life.

The female only effects of maternal TE contributions is reminiscent of the "W toxicity" hypothesis proposed by Peona et al. (2021). This hypothesis predicts that the accumulation of ERVs in the W chromosome would be detrimental to female survival. Our findings align with this hypothesis if we assume that the W chromosome drives maternal TE contributions as W-linked TE transcripts can represent up to 84.8% of all TE transcripts in female avian genomes (Warmuth et al., 2022). However,

we are not able to precisely determine the origin of the TE expansions we measured, but female biases in TE activity will most likely be related to the W chromosome. Additionally, our findings suggest that the detrimental association of W-linked TE activity in female offspring is observed when they are exposed to stress, making this association environmentally-dependent.

Sex-specific TE effects in LRS

We find that LRS is largely associated to TE dynamics in a sex specific manner. Higher maternal TE contributions were associated with increased male LRS, while higher paternal TE contributions had a marginally non-significant negative relationship. These associations suggest that male LRS is modulated by TE contributions mainly from the mother, although it is hard to determine why maternal contributions are beneficial and paternal contributions, detrimental.

Maternal TE contributions are more likely to be driven mostly by ERVs, since they have been accumulated in the W chromosome and are the most abundantly transcribed TEs in female avian genomes (Peona et al., 2021; Warmuth et al., 2022). Female LRS could remain the same despite different levels of parental TE contributions, as the presence of the ERV-rich W chromosome most likely overrides the effects of parental TE contributions and dominates the fraction of transcribed TEs (Warmuth et al., 2022). In contrast, paternal TE contributions could be driven by Z-linked TEs, which have a slightly higher insertion density than autosomal chromosomes due to reduced recombination (Olito & Abbott, 2025; Warmuth et al., 2022), likely dominated by mostly CR1 activity (Warmuth et al., 2022). The different outcome in LRS for maternal and paternal TE contributions could be linked to the different effects of novel insertions of ERV or CR1 elements, respectively.

One way in which TEs could contribute to higher LRS in males is by counteracting genome-wide homozygosity (De Kort et al., 2022), but the exact mechanism in which either maternal or paternal TE contributions could act on homozygosity is not straightforward. Understanding the potential role of TEs in overcoming inbreeding depression remains a promising field of research. This

is true even when considering results produced from analyses with a coarse scale as the one we present here.

Reproductive senescence in the Seychelles warbler

We did not find any significant associations between TE dynamics and reproductive senescence in the Seychelles warbler. This result decouples the effect of the interaction between maternal TE contributions and rainfall variance during early life on female offspring survival and reproductive senescence. This decoupling suggests that senescence trajectories in female offspring remain the same despite variability in maternal TE contributions.

However, we did find a male-specific association between maternal inbreeding and reproductive senescence patterns (Table 4). Male offspring from mother with higher inbreeding coefficients were shown to have an earlier onset and a higher rate of senescence (Figure 3). Intergenerational inbreeding effects in senescence have been explored by associating maternal homozygosity to offspring telomere length in the Seychelles warbler (Bebbington et al., 2016). Our association between maternal inbreeding and male offspring senescence resembles the findings by Bebbington et al. (2022) and Evans et al. (2026), who find a reduction in telomere length associated to higher maternal inbreeding coefficients in the Seychelles warbler and in the purple-crowned fairy-wrens (*Malurus coronatus coronatus*), respectively. In turn, shorter telomere lengths have been associated to earlier senescence (Chik et al., 2025; Huang et al., 2025; Pańczyszyn et al., 2020).

Inbreeding depression

Our models confirmed individual inbreeding associated to reduced survival and LRS. These findings have also been found in the Seychelles warbler in recent studies (Lee, et al., 2026b). In Pinto et al. (Pinto et al., 2026), an interaction between maternal inbreeding and the presence of siblings is associated to reduced offspring survival. We find maternal inbreeding associated to reduced offspring survival as reported before for the Seychelles warbler (Richardson et al., 2004), although we did not

test for interaction with sibling presence. Our models also fail to find any association between maternal inbreeding and LRS, as in the study by Pinto et al. (2026). This could be due to different model structures and different sample sizes.

Future directions

Our study faces large limitations given by the low coverage in our whole genome sequencing data. However, we have opted to use an appropriate approach for low coverage data which sacrifices insertion specificity, but allows us to estimate the TE content of a large amount of individuals (Goubert et al., 2015). Further studies could leverage long-read sequencing and transcriptomics to better describe the TE dynamics in the Seychelles warbler (Betancourt et al., 2024; Lanciano & Cristofari, 2020; Weissensteiner et al., 2020). A finer resolution of TE variants could also lead to studies where selective sweeps could be identified around TE insertions that boost fitness and provide more specific mechanisms of adaptation and evolvability (Riederer et al., 2022; Villanueva-Cañas et al., 2017). Finally, our study is correlational and would require further experimental work to understand the interplay between individual TE activity, intergenerational TE dynamics and inbreeding to assess causal relationships between TEs and fitness in a context of conservation concern.

Conclusions

In this study, we assessed the relationship between TE dynamics and survival, LRS and reproductive senescence in the Seychelles warbler (González-Mollinedo et al., Chapter 2). We found evidence that aligns with the hypothesis of “W toxicity” (Peona et al., 2021), but find that the association between higher maternal TE contributions and reduced survival to be mediated by the exposure to higher rainfall variance during early life. In contrast, larger maternal TE contributions were associated to higher LRS for male offspring, while paternal TE contributions were marginally associated to decreased LRS in male offspring, revealing a fitness trade-off around maternal TE contributions depending on offspring sex. No association between TE dynamics and reproductive senescence were observed, although maternal inbreeding was associated to accelerated male offspring senescence. These

findings constitute evidence of sex-specific fitness trade-offs associated to TE activity, contrasting with inbreeding depression in the Seychelles warbler.

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Supplementary Material

Survival

A main model was run to test for sex-specific interactions in survival using the *coxme* package (Table 2 main text). The following figure summarizes the Cox hazard ratios found in this including significant interactions later investigated in sex-specific models (Figure S1).

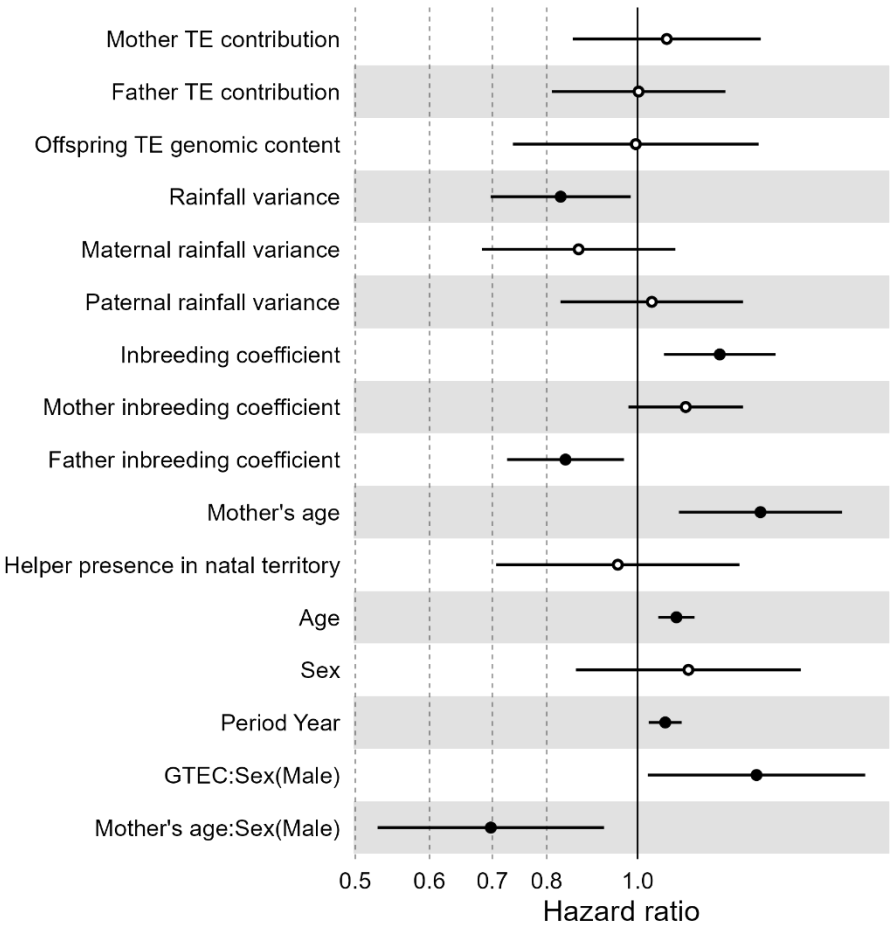


Figure S1. Hazard ratios and their respective confidence intervals (CIs) obtained from our Cox proportional hazards models with mixed effects testing for the effect of transposable element (TE) dynamics and other factors on the survival of the Seychelles warbler. Point estimate of the hazard ratios for each variable is shown as the point in the middle of the CIs, and significant hazards are shown as filled with black and no overlap of CIs with the vertical solid line.

We ran post-hoc sex-specific survival models to resolve the significant interactions we found.

The summary tables are included below detailing regression coefficients, hazards ratio, standard error (SE), Z-values and p-values (Table S1, S2)

Table S1. Summary of our male Cox proportional hazards mixed effect model of the effect of transposable element (TE) dynamics on survival of Seychelles warblers. Factors considered in the model are shown with their estimated coefficient, hazard ratio and hazard ratio standard error (SE), which determine the level of significance of each factor. Statistically significant hazard ratios have standard errors that do not overlap with 1 (p-values < 0.05), shown in bold. Regression coefficients, hazard ratios, standard errors (SE), Z-values and p-values are reported. Hazard ratios < 1 indicate a factor that improves survival, while a hazard ratio > 1 indicates a decrease in survival probability.

Factor	Coefficient	Hazard Ratio	SE	Z-value	P-value
Mother TE contribution	0.056	1.058	0.177	0.320	0.750
Father TE contribution	0.050	1.051	0.175	0.280	0.780
Offspring genomic TE content (GTEC)	0.275	1.316	0.196	1.400	0.160
Rainfall variance	-0.300	0.741	0.118	-2.530	0.011
Maternal rainfall variance	-0.290	0.748	0.177	-1.640	0.100
Paternal rainfall variance	0.078	1.081	0.149	0.520	0.600
Inbreeding coefficient	0.194	1.214	0.093	2.100	0.036
Mother inbreeding coefficient	0.040	1.041	0.101	0.390	0.690
Father inbreeding coefficient	-0.253	0.776	0.115	-2.190	0.028
Mother's age	-0.072	0.931	0.106	-0.680	0.500
Helper presence in natal territory	-0.188	0.829	0.227	-0.830	0.410
Age	0.128	1.136	0.034	3.730	0.000
Period Year	0.049	1.050	0.024	2.030	0.042
Random effects	Variable	Levels	SD	Variance	
Period year	Intercept	25.000	0.394	0.155	
Bird ID	Intercept	149.000	0.589	0.347	
Mother ID	Intercept	107.000	0.051	0.003	
Father ID	Intercept	101.000	0.020	0.000	

Table S2. Summary of our female Cox proportional hazards mixed effect model of the effect of transposable element (TE) dynamics on survival of Seychelles warblers. Factors considered in the model are shown with their estimated coefficient, hazard ratio and hazard ratio standard error (SE), which determine the level of significance of each factor. Statistically significant hazard ratios have standard errors that do not overlap with 1 (p-values < 0.05), shown in bold. Regression coefficients, hazard ratios, standard errors (SE), Z-values and p-values are reported. Hazard ratios < 1 indicate a factor that improves survival, while a hazard ratio > 1 indicates a decrease in survival probability.

Factor	Coefficient	Hazard Ratio	SE	Z-value	P-value
Mother TE contribution	0.283	1.327	0.158	1.790	0.074
Father TE contribution	-0.072	0.930	0.145	-0.500	0.620

Offspring genomic TE content (GTEC)	-0.055	0.947	0.182	-0.300	0.760
Rainfall variance	-0.240	0.787	0.119	-2.020	0.044
Maternal rainfall variance	0.031	1.032	0.166	0.190	0.850
Paternal rainfall variance	-0.041	0.960	0.181	-0.230	0.820
Inbreeding coefficient	0.232	1.261	0.113	2.050	0.041
Mother inbreeding coefficient	0.218	1.244	0.095	2.290	0.022
Father inbreeding coefficient	-0.095	0.909	0.095	-1.000	0.320
Mother's age	0.335	1.398	0.102	3.290	0.001
Helper presence in natal territory	0.091	1.095	0.204	0.450	0.660
Age	0.059	1.061	0.029	2.050	0.040
Period Year	0.103	1.108	0.027	3.750	<0.0001
Mother TE contribution:Rainfall variance	0.543	1.722	0.165	3.290	0.001
Random effects	Variable	Levels	SD	Variance	
Period year	Intercept	25	0.459	0.211	
Bird ID	Intercept	142	0.161	0.026	
Mother ID	Intercept	101	0.018	0.000	
Father ID	Intercept	101	0.370	0.137	

